

PharmGen AI — The Platform for AI-Designed Personalized Pharmaceuticals

“Your DNA. Your disease. Your drug. Made in hours, not decades.”

The Billion-Dollar Problem

The pharmaceutical industry is fundamentally broken.

- **\$2.6B average cost** to develop a single new drug
- **12-15 years** from discovery to patient
- **90% failure rate** in clinical trials
- **One-size-fits-all** drugs that work for only 30-50% of patients
- **30% of prescriptions** are ineffective for the individual patient
- **\$528B wasted annually** on drugs that don't work for specific patients

The dirty secret of modern medicine: We're still prescribing the same dose of the same drug to everyone, despite knowing that genetics, microbiome, lifestyle, and disease variants make drug response wildly individual.

The result: - Patients cycle through 3-5 medications before finding one that works - Adverse drug reactions kill 128,000 Americans annually - Cancer patients get “standard of care” while their tumors have unique mutations - Rare disease patients wait decades — or forever — for treatments

The technology convergence of 2026 changes everything: - AI can now design novel molecules in hours - 3D pharmaceutical printing enables on-demand drug manufacturing - Genomics costs dropped to \$100 per full sequence - Regulatory frameworks for personalized medicine are maturing - mRNA and peptide therapies can be customized rapidly

The missing piece: A unified platform that connects AI drug design → personalized formulation → on-demand manufacturing → patient delivery.

PharmGen AI is that platform.

The Vision

PharmGen AI creates the world's first **end-to-end personalized pharmaceutical platform** — where AI designs drugs optimized for YOUR specific biology, manufactured on-demand, and delivered to your doorstep.

We're building the AWS for personalized medicine.

The Three Revolutions We Enable

- 1. AI-Designed Drugs** - Novel molecules designed for your specific genetic variants - Optimized for your metabolic pathways - Targeted to your unique disease profile
- 2. On-Demand Manufacturing** - 3D printed pills with precise dosing - Compounded formulations in hours - Distributed manufacturing at scale

3. Continuous Optimization - Real-time monitoring and adjustment - Closed-loop feedback from wearables and biomarkers - Your drug evolves as your condition changes

How It Works

Platform Architecture

PHARMGEN AI PLATFORM

PATIENT PROFILE ENGINE	AI DRUG DESIGN ENGINE	PHARMA PRINT NETWORK
• Genomics • Proteomics • Microbiome • Phenotype • History	• Molecule Design • Formulate • Simulate • Optimize	• 3D Print • Compound • QC/QA • Package • Ship

CONTINUOUS OPTIMIZATION LOOP

Wearables → Biomarkers → Outcomes → Reformulation

1. Patient Profile Engine

Comprehensive biological fingerprinting:

- **Genomics:** Full genome sequencing + pharmacogenomic variants
- **Proteomics:** Protein expression profiles
- **Metabolomics:** Metabolic pathway analysis
- **Microbiome:** Gut bacteria composition (affects drug metabolism)
- **Phenotype:** Current health status, comorbidities, lifestyle
- **Drug history:** What's worked, what hasn't, adverse reactions

The output: A complete computational model of how YOUR body processes drugs.

2. AI Drug Design Engine

MoleculeForge: Our proprietary drug design AI

- **Foundation models** trained on 2B+ molecular structures
- **Generative chemistry:** Creates novel molecules optimized for your biology
- **Digital twins:** Simulates drug behavior in your specific body model
- **Multi-objective optimization:** Efficacy, safety, manufacturability, cost
- **Explainable AI:** Every design decision is traceable and auditable

Design modes:

Mode	Use Case	Timeline
Personalized Dose	Existing drug, optimized dosing for you	2 hours
Personalized Formulation	Modified delivery, timing, combinations	8 hours
Personalized Analog	Modified molecule for your variants	24-72 hours
Novel Molecule	New drug designed for your specific condition	1-4 weeks

3. PharmaPrint Network

Distributed on-demand manufacturing:

- **3D pharmaceutical printers** in 500+ licensed facilities
- **Automated compounding** for complex formulations
- **Real-time quality control** with AI vision and spectroscopy
- **Regulatory-compliant** manufacturing (FDA 503B, cGMP)
- **Same-day delivery** in major metros

Manufacturing capabilities:

- Pills and tablets (any shape, multi-layer, timed-release)
- Capsules (custom combinations, precision dosing)
- Topicals and transdermal patches
- Oral solutions and suspensions
- Peptides and biologics (partner network)
- mRNA therapeutics (partner network)

4. Continuous Optimization Loop

Your drug evolves with you:

- **Wearable integration:** Continuous glucose, HRV, sleep, activity
- **Biomarker tracking:** Regular blood panels, imaging, functional tests
- **Outcome monitoring:** Symptom tracking, quality of life scores
- **AI analysis:** Detects when reformulation would improve outcomes
- **Automatic adjustment:** New formulation generated and shipped

Target Markets

Market 1: Oncology Personalization

TAM: \$180B global cancer drug market

The problem: Cancer drugs are still prescribed by tumor type, not tumor genetics. Two patients with “breast cancer” may have completely different mutations.

PharmGen solution: - Sequence the tumor - Identify targetable mutations - Design combination therapy optimized for that specific cancer - Adjust in real-time based on tumor response

Entry point: Partner with comprehensive cancer centers for “precision oncology 2.0”

Market 2: Chronic Disease Optimization

TAM: \$420B chronic disease pharmaceutical market

Top opportunities:

Condition	Problem	PharmGen Value
Diabetes	Highly variable drug response	Personalized GLP-1 dosing, glucose-responsive formulations
Hypertension	50% of patients uncontrolled	Multi-drug combinations optimized for genotype
Depression	30% non-responders	Pharmacogenomic-guided antidepressant design
Autoimmune	Complex, individual disease	Personalized immunomodulators
Pain	Opioid crisis, poor alternatives	Non-opioid pain compounds for specific pain types

Market 3: Rare Diseases

TAM: \$260B rare disease market (fastest growing)

The tragedy: 7,000+ rare diseases, only 5% have approved treatments. Most patients are “orphaned.”

PharmGen revolution: - AI designs drugs for N=1 patients - Manufacturing scales down to single-patient batches - Regulatory pathway for personalized rare disease drugs - Patients who waited decades finally get treatments

Business model: Rare disease patients are willing to pay premium prices; insurers increasingly cover personalized treatments.

Market 4: Precision Compounding

TAM: \$15B compounding pharmacy market

Immediate revenue opportunity: - Existing compounding pharmacies lack AI optimization - PharmGen provides the intelligence layer - White-label platform for compounding networks

Market 5: Pharma R&D Acceleration

TAM: \$250B pharmaceutical R&D spending

Enterprise offering: - Pharma companies use PharmGen to design better clinical trial candidates
- Faster iteration cycles - Higher success rates - Patient stratification for trials

Competitive Landscape

Competitor	Approach	PharmGen Advantage
Traditional Pharma	Mass-market drugs	We personalize; they generalize
23andMe/Color	Genetic testing only	We go from data to delivered drug
Recursion/Insilico	AI drug discovery	We also manufacture and deliver
Compounding Pharmacies	Manual formulation	We're AI-native and scalable
Personalis/Tempus	Cancer genomics	We close the loop to treatment

Our moat: 1. **Vertical integration:** Only platform from design to delivery 2. **Data flywheel:** Every patient outcome improves our AI 3. **Manufacturing network:** Capital-intensive barrier to entry 4. **Regulatory expertise:** Navigating FDA personalized medicine frameworks 5. **IP portfolio:** Novel molecules designed by our AI

Business Model

Revenue Streams

1. Platform Subscriptions (B2B) - Healthcare systems: \$500K-\$5M/year - Pharma companies: \$1M-\$10M/year - Compounding networks: \$100K-\$500K/year

2. Per-Patient Fees (B2B2C) - Design fee: \$500-\$2,500 per personalized drug design - Manufacturing: \$50-\$500 per order - Monitoring: \$100/month continuous optimization

3. Direct-to-Patient (D2C) - Premium tier: \$299/month for continuous optimization - Chronic disease packages: \$149-\$499/month - One-time designs: \$999-\$4,999

4. Outcomes-Based Contracts - Shared savings with payers when personalized drugs outperform standard care - Risk-sharing with pharma partners

Unit Economics

Metric	Value
Average design cost	\$150
Average design price	\$1,200
Gross margin on design	87.5%
Average manufacturing cost	\$25
Average manufacturing price	\$150

Metric	Value
Gross margin on manufacturing	83.3%
Customer acquisition cost (B2B)	\$15,000
Customer lifetime value (B2B)	\$450,000
LTV:CAC	30:1

Go-to-Market Strategy

Phase 1: Oncology Beachhead (Year 1)

- Partner with 10 NCI-designated cancer centers
- Focus on treatment-resistant cancers
- Build clinical evidence base
- Establish regulatory pathway with FDA

Phase 2: Chronic Disease Expansion (Year 2)

- Launch diabetes personalization program
- Partner with large health systems (Kaiser, Geisinger)
- Integrate with major EHR systems
- Begin payer negotiations

Phase 3: Platform Scale (Year 3)

- White-label platform for compounding networks
- Pharma R&D partnerships
- International expansion (UK, EU, Japan)
- D2C premium launch

Phase 4: Full Ecosystem (Year 4+)

- Rare disease program at scale
 - Preventive medicine applications
 - Longevity optimization
 - Veterinary medicine expansion
-

Regulatory Strategy

FDA Pathway

Current regulatory assets: 1. **21st Century Cures Act:** Expedited pathways for personalized medicine 2. **FDA Guidance on Individualized Antisense Oligonucleotides:** Precedent for N=1 drugs 3. **503B Compounding:** Allows personalized formulations under physician supervision 4. **Breakthrough Therapy Designation:** Fast-track for serious conditions 5. **Right to Try Act:** Access for terminal patients

Our approach: - Start in 503B compounding (established regulatory framework) - Build evidence for FDA pre-submission meetings - Pursue breakthrough designation for oncology applications - Work with FDA on adaptive regulatory framework for AI-designed drugs

Compliance Infrastructure

- Real-time audit trails
 - Explainable AI for every drug design
 - cGMP manufacturing standards
 - HIPAA-compliant patient data handling
 - SOC 2 Type II certification
-

Technology Deep Dive

MoleculeForge AI Architecture

Foundation Layer: - Transformer models trained on 2B+ molecules (ChEMBL, PubChem, proprietary) - Protein structure prediction (AlphaFold-derived) - Pharmacokinetic modeling - Toxicity prediction (90%+ accuracy)

Personalization Layer: - Patient digital twin construction - Drug-body interaction simulation - Multi-objective optimization - Uncertainty quantification

Safety Layer: - Red-team adversarial testing - Human-in-the-loop verification - Regulatory compliance checking - Manufacturing feasibility validation

Manufacturing Technology

PharmaPrint 3D System: - Proprietary pharmaceutical printer design - Sub-milligram precision dosing - Multi-material capability (active + excipients) - Integrated QC (Raman spectroscopy, vision) - Throughput: 1,000 personalized doses/day per unit

Financial Projections

Metric	Year 1	Year 2	Year 3	Year 4	Year 5
Revenue	\$8M	\$35M	\$120M	\$350M	\$800M
Patients Served	5K	25K	100K	350K	900K
Healthcare Partners	15	75	250	600	1,200
Gross Margin	65%	72%	78%	82%	85%
EBITDA	-\$25M	-\$18M	\$5M	\$65M	\$180M

Path to \$1B revenue: Year 6 at current growth trajectory

Team Requirements

Founding Team Needs

CEO: Healthcare + tech background, regulatory navigation experience **CTO:** AI/ML in drug discovery, computational chemistry **CMO:** Practicing physician, precision medicine experience **CSO:** Pharmaceutical R&D, formulation science **COO:** Manufacturing operations, supply chain at scale

Key Hires (Year 1)

- Computational chemists (5)
 - ML engineers (8)
 - Regulatory affairs specialists (3)
 - Pharmaceutical manufacturing engineers (4)
 - Clinical operations (3)
 - Sales/BD for healthcare systems (4)
-

Funding Strategy

Seed Round: \$8M

- Build core AI platform
- Establish first manufacturing partnerships
- Regulatory pathway development
- Initial cancer center partnerships

Series A: \$35M

- Scale manufacturing network
- Launch chronic disease programs
- Expand clinical evidence
- Regulatory submissions

Series B: \$100M

- National healthcare system partnerships
- Pharma R&D platform
- International expansion
- D2C platform development

Target Investors

- Healthcare-focused VCs: a16z Bio, Andreessen Bio, GV
 - Deep tech investors: Lux Capital, Founders Fund
 - Strategic: Major health systems, pharma companies
 - Crossover: Fidelity, T. Rowe Price (later stages)
-

Risk Factors & Mitigation

Risk	Likelihood	Impact	Mitigation
Regulatory rejection	Medium	High	Start in established compounding framework
Manufacturing quality	Medium	High	Rigorous QC, redundant suppliers
AI design failures	Low	High	Human verification, conservative deployment
Payer resistance	Medium	Medium	Outcomes-based contracting, cost savings data
Competition from big pharma	Medium	Medium	Speed advantage, vertical integration
Patient adoption	Low	Medium	Physician-first go-to-market

Why Now?

Five forces converging in 2026:

1. **AI maturity:** Drug design AI has crossed the accuracy threshold
2. **Manufacturing tech:** 3D pharma printing is production-ready
3. **Genomics costs:** \$100 genomes make personalization economical
4. **Regulatory openness:** FDA actively encouraging precision medicine
5. **Market demand:** Patients demanding better than one-size-fits-all

The window is open. First mover will build the data moat.

The Vision: 2030

“In 2030, no one will take a mass-produced pill. Every medication will be designed for your unique biology, manufactured on-demand, and continuously optimized based on your real-time health data. PharmGen AI made this future possible.”

We’re not building a company. We’re building the future of medicine.

Call to Action

PharmGen AI is raising an **\$8M seed round** to build the world’s first AI-to-doorstep personalized pharmaceutical platform.

What we’re looking for: - Investors who understand healthcare transformation - Strategic partners in oncology and chronic disease - World-class talent in AI, drug discovery, and manufacturing

Contact: founders@pharmgen.ai

“The best time to personalize medicine was 20 years ago. The second best time is now.”

PharmGen AI — Your drug. Designed for you. Delivered to you.